Classification overview, ranks are significant with alpha=0.05

	gcn	gat	mpnn	attentivefp	svm-wwl	svm_rdkit	rf_rdkit	xgb_rdkit	dnn_rdkit	svm_minimol	rf_minimol	xgb_minimol	dnn_minimol	svm_moe-neutral	rf_moe-neutral	xgb_moe-neutral	dnn_moe-neutral
Ranks	9.48	11.08	14.71	11.55	10.0	9.36	5.88	7.11	9.04	8.28	8.54	10.19	9.49	8.93	4.5	6.17	8.69
bace	0.901 ± 0.018	0.886 ± 0.021	0.853 ± 0.039	0.875 ± 0.022	0.877 ± 0.016	0.886 ± 0.024	0.885 ± 0.028	0.881 ± 0.02	0.879 ± 0.021	0.894 ± 0.012	0.88 ± 0.019	0.887 ± 0.02	0.898 ± 0.018	0.861 ± 0.028	0.881 ± 0.025	0.88 ± 0.021	0.869 ± 0.022
bbbp	0.896 ± 0.027	0.897 ± 0.027	0.875 ± 0.037	0.895 ± 0.027	0.906 ± 0.023	0.921 ± 0.023	0.923 ± 0.022	0.923 ± 0.026	0.913 ± 0.025	0.915 ± 0.025	0.906 ± 0.028	0.912 ± 0.023	0.91 ± 0.023	0.902 ± 0.026	0.925 ± 0.021	0.92 ± 0.02	0.921 ± 0.026
clintox	0.876 ± 0.07	0.855 ± 0.069	0.851 ± 0.067	0.901 ± 0.03	0.758 ± 0.062	0.852 ± 0.076	0.864 ± 0.059	0.876 ± 0.046	0.845 ± 0.089	0.679 ± 0.168	0.77 ± 0.07	0.785 ± 0.057	0.832 ± 0.067	0.776 ± 0.065	0.795 ± 0.067	0.772 ± 0.074	0.782 ± 0.069
sider	0.624 ± 0.018	0.624 ± 0.023	0.571 ± 0.029	0.628 ± 0.03	0.642 ± 0.09	0.629 ± 0.082	0.654 ± 0.081	0.632 ± 0.086	0.63 ± 0.035	0.639 ± 0.103	0.654 ± 0.076	0.643 ± 0.084	0.628 ± 0.026	0.636 ± 0.086	0.663 ± 0.082	0.656 ± 0.085	0.623 ± 0.032
tox21	0.83 ± 0.014	0.833 ± 0.017	0.814 ± 0.02	0.848 ± 0.014	na	0.813 ± 0.065	0.837 ± 0.062	0.829 ± 0.068	0.837 ± 0.015	0.858 ± 0.066	0.863 ± 0.059	0.86 ± 0.067	0.865 ± 0.017	0.82 ± 0.065	0.841 ± 0.057	0.836 ± 0.067	0.838 ± 0.013
clintox1	0.907 ± 0.051	0.883 ± 0.053	0.772 ± 0.126	0.891 ± 0.05	0.764 ± 0.05	0.856 ± 0.072	0.849 ± 0.072	0.863 ± 0.046	0.864 ± 0.069	0.468 ± 0.136	0.773 ± 0.04	0.777 ± 0.067	0.838 ± 0.052	0.795 ± 0.054	0.772 ± 0.081	0.758 ± 0.098	0.764 ± 0.102
clintox2	0.905 ± 0.047	0.863 ± 0.076	0.879 ± 0.036	0.912 ± 0.043	0.748 ± 0.073	0.846 ± 0.088	0.875 ± 0.051	0.864 ± 0.049	0.837 ± 0.09	0.783 ± 0.065	0.799 ± 0.05	0.769 ± 0.053	0.826 ± 0.048	0.804 ± 0.047	0.816 ± 0.048	0.797 ± 0.058	0.778 ± 0.068
sider1	0.655 ± 0.039	0.64 ± 0.061	0.628 ± 0.058	0.632 ± 0.045	0.701 ± 0.044	0.713 ± 0.036	0.679 ± 0.043	0.68 ± 0.047	0.698 ± 0.049	0.719 ± 0.041	0.649 ± 0.041	0.721 ± 0.042	0.709 ± 0.039	0.703 ± 0.04	0.74 ± 0.042	0.721 ± 0.042	0.693 ± 0.031
sider2	0.583 ± 0.045	0.551 ± 0.088	0.54 ± 0.057	0.57 ± 0.059	0.607 ± 0.042	0.455 ± 0.096	0.65 ± 0.072	0.624 ± 0.056	0.596 ± 0.062	0.539 ± 0.091	0.625 ± 0.056	0.595 ± 0.044	0.626 ± 0.06	0.596 ± 0.047	0.662 ± 0.061	0.652 ± 0.052	0.616 ± 0.07
sider3	0.568 ± 0.202	0.546 ± 0.214	0.555 ± 0.178	0.459 ± 0.221	0.467 ± 0.109	0.484 ± 0.178	0.607 ± 0.153	0.646 ± 0.243	0.504 ± 0.142	0.52 ± 0.266	0.617 ± 0.166	0.645 ± 0.201	0.465 ± 0.195	0.606 ± 0.211	0.565 ± 0.221	0.468 ± 0.173	0.567 ± 0.232
sider4	0.623 ± 0.051	0.632 ± 0.027	0.594 ± 0.047	0.565 ± 0.045	0.622 ± 0.039	0.622 ± 0.039	0.634 ± 0.039	0.626 ± 0.043	0.614 ± 0.042	0.677 ± 0.045	0.605 ± 0.038	0.621 ± 0.03	0.625 ± 0.037	0.671 ± 0.051	0.633 ± 0.053	0.617 ± 0.051	0.637 ± 0.047
sider5	0.657 ± 0.06	0.648 ± 0.07	0.636 ± 0.059	0.643 ± 0.063	0.67 ± 0.076	0.645 ± 0.057	0.68 ± 0.07	0.656 ± 0.06	0.669 ± 0.067	0.671 ± 0.05	0.689 ± 0.055	0.675 ± 0.058	0.676 ± 0.049	0.68 ± 0.054	0.688 ± 0.064	0.692 ± 0.056	0.664 ± 0.062
sider6	0.605 ± 0.05	0.641 ± 0.044	0.548 ± 0.044	0.616 ± 0.046	0.646 ± 0.049	0.589 ± 0.031	0.653 ± 0.042	0.628 ± 0.048	0.634 ± 0.049	0.642 ± 0.038	0.674 ± 0.038	0.646 ± 0.048	0.654 ± 0.039	0.611 ± 0.055	0.673 ± 0.039	0.663 ± 0.034	0.629 ± 0.043
sider7	0.675 ± 0.095	0.676 ± 0.07	0.667 ± 0.089	0.627 ± 0.08	0.728 ± 0.083	0.702 ± 0.056	0.749 ± 0.074	0.753 ± 0.075	0.688 ± 0.102	0.726 ± 0.094	0.675 ± 0.087	0.671 ± 0.076	0.658 ± 0.101	0.719 ± 0.079	0.753 ± 0.068	0.722 ± 0.065	0.694 ± 0.07
sider8	0.59 ± 0.058	0.573 ± 0.049	0.551 ± 0.059	0.588 ± 0.044	0.635 ± 0.049	0.599 ± 0.038	0.646 ± 0.051	0.59 ± 0.04	0.622 ± 0.042	0.481 ± 0.063	0.604 ± 0.046	0.594 ± 0.054	0.607 ± 0.074	0.605 ± 0.059	0.645 ± 0.066	0.622 ± 0.067	0.617 ± 0.057
sider9	0.592 ± 0.065	0.574 ± 0.045	0.555 ± 0.044	0.576 ± 0.058	0.604 ± 0.037	0.587 ± 0.068	0.61 ± 0.06	0.596 ± 0.049	0.592 ± 0.072	0.606 ± 0.056	0.615 ± 0.045	0.601 ± 0.038	0.601 ± 0.046	0.57 ± 0.057	0.631 ± 0.057	0.603 ± 0.067	0.613 ± 0.05
sider10	0.655 ± 0.052	0.651 ± 0.052	0.647 ± 0.059	0.599 ± 0.04	0.698 ± 0.039	0.67 ± 0.04	0.69 ± 0.032	0.674 ± 0.033	0.664 ± 0.038	0.724 ± 0.035	0.689 ± 0.035	0.687 ± 0.034	0.68 ± 0.035	0.676 ± 0.037	0.721 ± 0.035	0.705 ± 0.041	0.673 ± 0.036
sider11	0.692 ± 0.042	0.697 ± 0.043	0.629 ± 0.045	0.639 ± 0.046	0.721 ± 0.039	0.684 ± 0.044	0.711 ± 0.04	0.704 ± 0.054	0.689 ± 0.058	0.74 ± 0.049	0.724 ± 0.062	0.715 ± 0.037	0.701 ± 0.041	0.711 ± 0.036	0.734 ± 0.032	0.693 ± 0.054	0.671 ± 0.027
sider12	0.624 ± 0.075	0.627 ± 0.076	0.535 ± 0.087	0.512 ± 0.075	0.582 ± 0.07	0.595 ± 0.105	0.575 ± 0.09	0.595 ± 0.097	0.584 ± 0.105	0.59 ± 0.075	0.606 ± 0.071	0.58 ± 0.084	0.615 ± 0.093	0.562 ± 0.089	0.611 ± 0.095	0.64 ± 0.069	0.565 ± 0.102
sider13	0.696 ± 0.053	0.655 ± 0.055	0.678 ± 0.037	0.638 ± 0.062	0.701 ± 0.033	0.697 ± 0.038	0.704 ± 0.028	0.682 ± 0.054	0.692 ± 0.036	0.697 ± 0.041	0.706 ± 0.045	0.7 ± 0.064	0.701 ± 0.053	0.683 ± 0.054	0.726 ± 0.046	0.705 ± 0.061	0.693 ± 0.045
sider14	0.547 ± 0.073	0.555 ± 0.074	0.529 ± 0.054	0.532 ± 0.085	0.605 ± 0.059	0.595 ± 0.103	0.571 ± 0.051	0.564 ± 0.075	0.576 ± 0.072	0.589 ± 0.08	0.56 ± 0.067	0.547 ± 0.051	0.576 ± 0.068	0.597 ± 0.071	0.629 ± 0.065	0.603 ± 0.065	0.603 ± 0.066
sider15	0.64 ± 0.053	0.631 ± 0.076	0.61 ± 0.062	0.585 ± 0.046	0.679 ± 0.062	0.639 ± 0.061	0.642 ± 0.049	0.656 ± 0.053	0.65 ± 0.06	0.634 ± 0.049	0.679 ± 0.044	0.65 ± 0.046	0.637 ± 0.047	0.632 ± 0.06	0.691 ± 0.061	0.669 ± 0.056	0.645 ± 0.052
sider16	0.681 ± 0.046	0.693 ± 0.057	0.645 ± 0.067	0.688 ± 0.06	0.726 ± 0.047	0.71 ± 0.051	0.709 ± 0.041	0.708 ± 0.044	0.7 ± 0.04	0.724 ± 0.049	0.731 ± 0.034	0.718 ± 0.037	0.73 ± 0.053	0.721 ± 0.044	0.744 ± 0.047	0.715 ± 0.039	0.716 ± 0.045
sider17	0.641 ± 0.049	0.581 ± 0.103	0.59 ± 0.132	0.662 ± 0.098	0.61 ± 0.076	0.576 ± 0.099	0.611 ± 0.09	0.61 ± 0.083	0.673 ± 0.113	0.606 ± 0.124	0.678 ± 0.055	0.666 ± 0.069	0.662 ± 0.065	0.573 ± 0.066	0.616 ± 0.056	0.607 ± 0.064	0.663 ± 0.098
sider18	0.596 ± 0.062	0.561 ± 0.037	0.51 ± 0.066	0.556 ± 0.041	0.563 ± 0.074	0.589 ± 0.065	0.602 ± 0.063	0.604 ± 0.06	0.61 ± 0.051	0.602 ± 0.056	0.577 ± 0.037	0.587 ± 0.048	0.613 ± 0.06	0.628 ± 0.067	0.585 ± 0.064	0.583 ± 0.06	0.573 ± 0.066
sider19	0.63 ± 0.046	0.605 ± 0.065	0.585 ± 0.058	0.588 ± 0.063	0.677 ± 0.066	0.651 ± 0.058	0.67 ± 0.065	0.658 ± 0.06	0.643 ± 0.063	0.655 ± 0.049	0.646 ± 0.053	0.632 ± 0.065	0.654 ± 0.053	0.629 ± 0.067	0.668 ± 0.06	0.663 ± 0.055	0.653 ± 0.048
sider20	0.613 ± 0.07	0.548 ± 0.044	0.591 ± 0.041	0.535 ± 0.039	0.616 ± 0.072	0.624 ± 0.043	0.61 ± 0.075	0.622 ± 0.084	0.623 ± 0.059	0.654 ± 0.055	0.628 ± 0.064	0.596 ± 0.062	0.603 ± 0.062	0.602 ± 0.06	0.625 ± 0.078	0.648 ± 0.065	0.634 ± 0.064
sider21	0.65 ± 0.046	0.656 ± 0.038	0.608 ± 0.053	0.602 ± 0.059	0.698 ± 0.042	0.646 ± 0.033	0.69 ± 0.053	0.663 ± 0.056	0.655 ± 0.054	0.679 ± 0.046	0.685 ± 0.044	0.648 ± 0.056	0.667 ± 0.048	0.688 ± 0.037	0.698 ± 0.047	0.672 ± 0.058	0.661 ± 0.036
sider22	0.647 ± 0.037	0.635 ± 0.049	0.585 ± 0.061	0.613 ± 0.051	0.666 ± 0.055	0.646 ± 0.044	0.638 ± 0.042	0.658 ± 0.037	0.632 ± 0.042	0.656 ± 0.049	0.664 ± 0.042	0.651 ± 0.056	0.647 ± 0.043	0.631 ± 0.049	0.644 ± 0.034	0.648 ± 0.04	0.655 ± 0.046
sider23	0.593 ± 0.087	0.593 ± 0.073	0.562 ± 0.101	0.555 ± 0.094	0.566 ± 0.07	0.548 ± 0.071	0.598 ± 0.08	0.557 ± 0.063	0.597 ± 0.066	0.59 ± 0.073	0.631 ± 0.069	0.579 ± 0.06	0.578 ± 0.07	0.596 ± 0.046	0.613 ± 0.058	0.572 ± 0.084	0.583 ± 0.061
sider24	0.638 ± 0.041	0.602 ± 0.043	0.595 ± 0.026	0.544 ± 0.07	0.641 ± 0.037	0.659 ± 0.033	0.664 ± 0.023	0.642 ± 0.037	0.636 ± 0.052	0.653 ± 0.024	0.616 ± 0.029	0.64 ± 0.035	0.649 ± 0.046	0.661 ± 0.034	0.66 ± 0.04	0.634 ± 0.048	0.633 ± 0.034
sider25	0.696 ± 0.054	0.653 ± 0.062	0.624 ± 0.063	0.611 ± 0.065	0.698 ± 0.054	0.689 ± 0.041	0.71 ± 0.06	0.686 ± 0.066	0.656 ± 0.055	0.744 ± 0.054	0.704 ± 0.062	0.675 ± 0.06	0.685 ± 0.049	0.722 ± 0.068	0.735 ± 0.061	0.733 ± 0.048	0.676 ± 0.031
sider26	0.68 ± 0.06	0.662 ± 0.102	0.58 ± 0.094	0.527 ± 0.124	0.639 ± 0.059	0.608 ± 0.101	0.696 ± 0.072	0.675 ± 0.072	0.673 ± 0.064	0.499 ± 0.107	0.657 ± 0.105	0.672 ± 0.082	0.677 ± 0.071	0.644 ± 0.096	0.669 ± 0.088	0.706 ± 0.055	0.665 ± 0.079
sider27	0.556 ± 0.041	0.534 ± 0.061	0.539 ± 0.04	0.534 ± 0.046	0.571 ± 0.038	0.599 ± 0.049	0.607 ± 0.05	0.6 ± 0.049	0.583 ± 0.037	0.63 ± 0.031	0.601 ± 0.052	0.58 ± 0.044	0.572 ± 0.043	0.592 ± 0.055	0.638 ± 0.043	0.599 ± 0.047	0.59 ± 0.051
tox211	0.779 ± 0.048	0.79 ± 0.062	0.771 ± 0.068	0.799 ± 0.052	0.778 ± 0.052	0.791 ± 0.048	0.815 ± 0.053	0.792 ± 0.062	0.799 ± 0.055	0.691 ± 0.154	0.698 ± 0.159	0.701 ± 0.144	0.686 ± 0.171	0.776 ± 0.056	0.781 ± 0.062	0.793 ± 0.053	0.796 ± 0.067
tox212	0.839 ± 0.065	0.848 ± 0.078	0.803 ± 0.079	0.852 ± 0.056	0.835 ± 0.053	0.833 ± 0.042	0.852 ± 0.06	0.833 ± 0.054	0.822 ± 0.08	0.812 ± 0.124	0.737 ± 0.15	0.787 ± 0.129	0.749 ± 0.198	0.819 ± 0.046	0.851 ± 0.054	0.824 ± 0.059	0.833 ± 0.075
tox213	0.875 ± 0.022	0.88 ± 0.021	0.873 ± 0.022	0.889 ± 0.017	0.856 ± 0.019	0.883 ± 0.016	0.891 ± 0.019	0.898 ± 0.018	0.888 ± 0.018	0.844 ± 0.059	0.859 ± 0.058	0.845 ± 0.064	0.834 ± 0.071	0.885 ± 0.014	0.886 ± 0.02	0.897 ± 0.02	0.886 ± 0.018
tox214	0.839 ± 0.048	0.849 ± 0.028	0.772 ± 0.046	0.829 ± 0.037	0.833 ± 0.05	0.826 ± 0.041	0.861 ± 0.027	0.859 ± 0.031	0.835 ± 0.056	0.839 ± 0.072	0.777 ± 0.084	0.824 ± 0.087	0.81 ± 0.114	0.816 ± 0.042	0.868 ± 0.035	0.861 ± 0.038	0.859 ± 0.037
tox215	0.718 ± 0.026	0.729 ± 0.023	0.708 ± 0.017	0.734 ± 0.022	0.71 ± 0.036	0.74 ± 0.037	0.725 ± 0.037	0.726 ± 0.034	0.746 ± 0.023	0.575 ± 0.067	0.576 ± 0.049	0.555 ± 0.079	0.574 ± 0.048	0.739 ± 0.037	0.74 ± 0.026	0.737 ± 0.029	0.749 ± 0.019
tox216	0.815 ± 0.023	0.8 ± 0.026	0.759 ± 0.035	0.799 ± 0.038	0.781 ± 0.054	0.837 ± 0.039	0.825 ± 0.045	0.827 ± 0.041	0.804 ± 0.035	0.688 ± 0.108	0.734 ± 0.09	0.701 ± 0.094	0.699 ± 0.103	0.831 ± 0.037	0.831 ± 0.045	0.824 ± 0.038	0.797 ± 0.026
tox217	0.803 ± 0.074	0.791 ± 0.059	0.75 ± 0.054	0.789 ± 0.048	0.787 ± 0.061	0.808 ± 0.056	0.844 ± 0.059	0.848 ± 0.059	0.804 ± 0.066	0.85 ± 0.138	0.787 ± 0.202	0.769 ± 0.167	0.785 ± 0.201	0.864 ± 0.05	0.853 ± 0.074	0.835 ± 0.082	0.837 ± 0.051
tox218	0.801 ± 0.031	0.814 ± 0.029	0.785 ± 0.029	0.809 ± 0.026	0.785 ± 0.031	0.815 ± 0.017	0.831 ± 0.021	0.832 ± 0.023	0.819 ± 0.023	0.724 ± 0.047	0.748 ± 0.049	0.729 ± 0.068	0.729 ± 0.05	0.797 ± 0.02	0.82 ± 0.021	0.834 ± 0.025	0.818 ± 0.023
tox219	0.835 ± 0.04	0.823 ± 0.042	0.809 ± 0.045	0.822 ± 0.036	0.791 ± 0.03	0.825 ± 0.036	0.84 ± 0.025	0.837 ± 0.06	0.819 ± 0.043	0.961 ± 0.082	0.877 ± 0.131	0.86 ± 0.09	0.669 ± 0.26	0.799 ± 0.037	0.85 ± 0.036	0.845 ± 0.054	0.829 ± 0.042
tox2110	0.782 ± 0.035	0.78 ± 0.044	0.74 ± 0.043	0.803 ± 0.04	0.737 ± 0.034	0.801 ± 0.03	0.771 ± 0.042	0.787 ± 0.025	0.782 ± 0.048	0.379 ± 0.086	0.695 ± 0.112	0.609 ± 0.149	0.713 ± 0.082	0.802 ± 0.031	0.813 ± 0.023	0.798 ± 0.035	0.794 ± 0.039
tox2111	0.902 ± 0.02	0.9 ± 0.018	0.874 ± 0.015	0.907 ± 0.015	0.893 ± 0.019	0.923 ± 0.015	0.925 ± 0.015	0.936 ± 0.018	0.906 ± 0.018	0.91 ± 0.051	0.908 ± 0.048	0.89 ± 0.058	0.87 ± 0.08	0.92 ± 0.016	0.926 ± 0.013	0.936 ± 0.016	0.913 ± 0.018
tox2112	0.841 ± 0.036	0.832 ± 0.036	0.782 ± 0.038	0.85 ± 0.021	0.822 ± 0.03	0.821 ± 0.029	0.858 ± 0.027	0.86 ± 0.036	0.851 ± 0.037	0.792 ± 0.196	0.87 ± 0.098	0.827 ± 0.152	0.828 ± 0.16	0.815 ± 0.037	0.833 ± 0.022	0.852 ± 0.033	0.829 ± 0.05
hiv	0.835 ± 0.023	0.834 ± 0.026	0.807 ± 0.024	0.811 ± 0.024	na	0.792 ± 0.023	0.81 ± 0.021	0.806 ± 0.022	0.785 ± 0.026	0.858 ± 0.02	0.852 ± 0.027	0.847 ± 0.026	0.856 ± 0.025	0.806 ± 0.023	0.83 ± 0.025	0.819 ± 0.027	0.794 ± 0.023

Legend
- First
- Second
- Third
- WWL
- GNN
- RDKit
- Minimol